

PCAP™ Mock Examination

Certified Associate in Python Programming · Exam PCAP-31-03

Candidate Name:		Date:	
Candidate ID:		Score:	

INSTRUCTIONS TO CANDIDATES

- 1. This examination contains 40 questions divided into 5 sections.
- 2. Questions marked [Single Select] have exactly ONE correct answer.
- 3. Questions marked [Multiple Select] have TWO or more correct answers — mark ALL that apply.
- 4. Questions marked [Code Analysis] require you to trace code execution.
- 5. Each question is worth equal marks. There is no negative marking.
- 6. You have 65 minutes to complete the exam. Manage your time carefully.
- 7. A score of 70% or higher (28/40) is required to pass.
- 8. Circle or clearly mark your chosen answer(s) for each question.

Section	Topic	Weight	Questions
1	Modules and Packages	12%	6
2	Exceptions	14%	5
3	Strings	18%	8
4	Object-Oriented Programming	34%	12
5	Miscellaneous (List Comprehensions, Lambdas, Closures, I/O)	22%	9

Section 1: Modules and Packages

12% · 6 Questions

Q1

Single Select

Which statement correctly imports only the sqrt function from the math module?

- ☐ A. import math.sqrt
- ☐ B. from math import sqrt
- ☐ C. import sqrt from math
- ☐ D. include math.sqrt

Q2

Single Select

What does the sys.path variable contain?

- ☐ A. The Python version string
- ☐ B. A list of directories Python searches for modules
- ☐ C. The path to the Python executable
- ☐ D. A dictionary of installed packages

Q3

Single Select

What is the output of the following code?

```
import math
print(math.ceil(4.1), math.floor(4.9))
```

- ☐ A. 4 4
- ☐ B. 5 5
- ☐ C. 5 4
- ☐ D. 4 5

Q4

Single Select

Which file must be present in a directory for Python to treat it as a package?

- ☐ A. __main__.py
- ☐ B. __init__.py
- ☐ C. __package__.py
- ☐ D. __module__.py

Q5

Single Select

What does the dir() function return when called on a module?

- ☐ A. The module's source code
- ☐ B. A list of the module's attributes and functions
- ☐ C. The module's file path
- ☐ D. The module's version number

Q6

Multiple Select

Which of the following are valid import statements? (Select TWO)

- ☐ A. import os as operating_system
- ☐ B. from sys import *
- ☐ C. import from math
- ☐ D. include random

Section 2: Exceptions

14% · 5 Questions

Q7

Single Select

What is the correct order of exception handling clauses?

- ☐ A. try → finally → except → else
- ☐ B. try → except → else → finally
- ☐ C. try → else → except → finally
- ☐ D. try → except → finally → else

Q8

Single Select

What will the following code print?

```
try:
    x = 1 / 0
except ZeroDivisionError:
    print('zero')
else:
    print('ok')
finally:
    print('done')
```

- ☐ A. zero done
- ☐ B. ok done
- ☐ C. zero ok done
- ☐ D. done

Q9

Single Select

Which exception class is the base class for all built-in, non-system-exiting exceptions?

- ☐ A. BaseException
- ☐ B. Exception
- ☐ C. RuntimeError
- ☐ D. StandardError

Q10

Single Select

How do you define a custom exception called `NetworkError` that inherits from `Exception`?

- ☐ A. `class NetworkError extends Exception: pass`
- ☐ B. `class NetworkError(Exception): pass`
- ☐ C. `class NetworkError inherits Exception: pass`
- ☐ D. `exception NetworkError(Exception): pass`

Q11

Multiple Select

Which statements about the `assert` statement are correct? (Select TWO)

- ☐ A. `assert` raises `AssertionError` when the condition is `False`
- ☐ B. `assert` raises `ValueError` when the condition is `False`
- ☐ C. `assert` can include an optional message: `assert cond, 'msg'`
- ☐ D. `assert` always runs even in optimized mode (`-O` flag)

Section 3: Strings

18% · 8 Questions

Q12

Single Select

What is the output of the following?

```
s = 'Python'
print(s[1:4])
```

- ☐ A. `Pyt`
- ☐ B. `yth`
- ☐ C. `ytho`
- ☐ D. `ython`

Q13

Single Select

What does `ord('A')` return?

- ☐ A. 64
- ☐ B. 65
- ☐ C. 97
- ☐ D. 'A'

Q14

Single Select

What is the output of the following?

```
words = ['Hello', 'World']
print(' '.join(words))
```

- ☐ A. `HelloWorld`
- ☐ B. `Hello World`
- ☐ C. `['Hello', 'World']`
- ☐ D. `Hello,World`

Q15

Single Select

What is the difference between `str.find()` and `str.index()`?

- ☐ A. `find()` is case-sensitive; `index()` is not
- ☐ B. `find()` returns -1 if not found; `index()` raises `ValueError`
- ☐ C. `index()` returns -1 if not found; `find()` raises `ValueError`
- ☐ D. There is no difference

Q16

Single Select

What encoding standard supports all Unicode characters using variable-length bytes?

- ☐ A. ASCII
- ☐ B. UTF-16
- ☐ C. UTF-8
- ☐ D. Latin-1

Q17

Single Select

What is the output of the following?

```
s = ' hello '  
print(s.strip())
```

- ☐ A. ' hello '
- ☐ B. 'hello'
- ☐ C. 'hello '
- ☐ D. ' hello'

Q18

Multiple Select

Which of the following string methods return `True` for the string `'Hello123'`? (Select TWO)

- ☐ A. `'Hello123'.isalpha()`
- ☐ B. `'Hello123'.isalnum()`
- ☐ C. `'Hello123'.isdigit()`
- ☐ D. `'Hello123'.isidentifier()`

Q19

Single Select

What is the output of the following?

```
print(sorted('dcba'))
```

- ☐ A. 'abcd'
- ☐ B. ['a', 'b', 'c', 'd']
- ☐ C. abcd
- ☐ D. ('a','b','c','d')

Section 4: Object-Oriented Programming

34% · 12 Questions

Q20

Single Select

What is the purpose of the `__init__` method in a Python class?

- ☐ A. It is called when the class is imported
- ☐ B. It initializes a new object instance
- ☐ C. It defines class-level variables only
- ☐ D. It is called when the object is deleted

Q21

Single Select

What will the following code print?

```
class Dog:
    species = 'Canine'
    def __init__(self, name):
        self.name = name

d = Dog('Rex')
print(d.species, d.name)
```

- ☐ A. Canine Rex
- ☐ B. Rex Canine
- ☐ C. Error
- ☐ D. None Rex

Q22

Single Select

Which of the following correctly defines a private attribute in Python?

- ☐ A. `private self.value = 10`
- ☐ B. `self._value = 10`
- ☐ C. `self.__value = 10`
- ☐ D. Both B and C are used for privacy, but C applies name mangling

Q23

Single Select

What is method overriding?

- ☐ A. Defining two methods with the same name in the same class
- ☐ B. Redefining a parent class method in a child class
- ☐ C. Calling a method with different argument types
- ☐ D. Deleting a method from a parent class

Q24

Single Select

What does `isinstance(obj, MyClass)` return?

- ☐ A. True only if `obj` is an exact instance of `MyClass`
- ☐ B. True if `obj` is an instance of `MyClass` or any subclass of it
- ☐ C. True if `MyClass` has an attribute matching `obj`
- ☐ D. The class name of `obj`

Q25

Code Analysis

What is the output of the following code?

```
class Animal:
    def speak(self):
        return 'Some sound'

class Cat(Animal):
    def speak(self):
        return 'Meow'

class Dog(Animal):
    def speak(self):
        return 'Woof'

for a in [Cat(), Dog(), Animal()]:
    print(a.speak())
```

- ☐ A. Some sound Some sound Some sound
- ☐ B. Meow Woof Some sound
- ☐ C. Meow Woof Meow
- ☐ D. Error

Q26

Single Select

What does the `__str__` method allow you to do?

- ☐ A. Convert an object to bytes
- ☐ B. Control the string representation returned by `str()` and `print()`
- ☐ C. Define how two objects are compared
- ☐ D. Restrict attribute access

Q27

Single Select

In multiple inheritance class `C(A, B)`, if both `A` and `B` define method `foo()`, which version does `C` use by default?

- ☐ A. B's version
- ☐ B. A's version
- ☐ C. Both versions are merged
- ☐ D. Python raises an error

Q28

Multiple Select

Which of the following are true about class variables? (Select TWO)

- ☐ A. They are shared across all instances of the class
- ☐ B. They are defined inside the `__init__` method
- ☐ C. They can be accessed via the class name or an instance
- ☐ D. Modifying them on one instance always affects all instances

Q29

Single Select

What does `hasattr(obj, 'name')` return?

- ☐ A. The value of `obj.name`
- ☐ B. True if `obj` has an attribute called 'name', False otherwise
- ☐ C. The type of `obj.name`
- ☐ D. True only if 'name' is a method

Q30

Code Analysis

What is printed by the following code?

```
class Counter:
    count = 0
    def __init__(self):
        Counter.count += 1

a = Counter()
b = Counter()
c = Counter()
print(Counter.count)
```

- ☐ A. 0
- ☐ B. 1
- ☐ C. 3
- ☐ D. Error

Q31

Single Select

What keyword is used to call the parent class's method from a child class?

- ☐ A. `parent()`
- ☐ B. `base()`
- ☐ C. `super()`
- ☐ D. `inherit()`

Section 5: Miscellaneous (List Comprehensions, Lambdas, Closures, I/O)

22% · 9 Questions

Q32

Single Select

What is the output of the following list comprehension?

```
result = [x**2 for x in range(5) if x % 2 == 0]
print(result)
```

- ☐ A. [1, 9, 25]
- ☐ B. [0, 4, 16]
- ☐ C. [0, 1, 4, 9, 16]
- ☐ D. [4, 16]

Q33

Single Select

What does the following lambda expression do?

```
f = lambda x, y: x if x > y else y
```

- ☐ A. Returns the sum of x and y
- ☐ B. Returns the minimum of x and y
- ☐ C. Returns the maximum of x and y
- ☐ D. Returns True if x > y

Q34

Single Select

What is the output of the following?

```
nums = [1, 2, 3, 4, 5]
result = list(filter(lambda x: x % 2 != 0, nums))
print(result)
```

- ☐ A. [2, 4]
- ☐ B. [1, 3, 5]
- ☐ C. [1, 2, 3, 4, 5]
- ☐ D. []

Q35

Code Analysis

What does the following closure print when called as add5(3)?

```
def make_adder(n):
    def adder(x):
        return x + n
    return adder

add5 = make_adder(5)
print(add5(3))
```

- ☐ A. 3
- ☐ B. 5
- ☐ C. 8
- ☐ D. Error

Q36

Single Select

Which file mode opens a file for writing and truncates it if it already exists?

- ☐ A. 'r'
- ☐ B. 'a'
- ☐ C. 'w'
- ☐ D. 'x'

Q37**Single Select**

What is the correct way to safely open and read a file in Python?

- ☐ A. `f = open('file.txt'); data = f.read()`
- ☐ B. `with open('file.txt', 'r') as f: data = f.read()`
- ☐ C. `open('file.txt') as f: data = f.read()`
- ☐ D. `file.open('file.txt', read=True)`

Q38**Multiple Select**

Which of the following are true about `map()`? (Select TWO)

- ☐ A. `map()` applies a function to every item in an iterable
- ☐ B. `map()` returns a list directly in Python 3
- ☐ C. `map()` returns a map object (iterator) in Python 3
- ☐ D. `map()` can only work with lambda functions

Q39**Single Select**

What is the output of the following?

```
result = list(map(lambda x: x * 2, [1, 2, 3]))
print(result)
```

- ☐ A. `[1, 2, 3]`
- ☐ B. `[2, 4, 6]`
- ☐ C. `[1, 4, 9]`
- ☐ D. `[3, 6, 9]`

Q40**Single Select**

Which method reads all lines of a file and returns them as a list of strings?

- ☐ A. `file.read()`
- ☐ B. `file.readline()`
- ☐ C. `file.readlines()`
- ☐ D. `file.readall()`

ANSWER KEY

For personal study use only. Review each explanation after marking your paper.

Q#	Answer	Q#	Answer	Q#	Answer	Q#	Answer
1	B	11	A, C	21	A	31	C
2	B	12	B	22	D	32	B
3	C	13	B	23	B	33	C
4	B	14	B	24	B	34	B
5	B	15	B	25	B	35	C
6	A, B	16	C	26	B	36	C
7	B	17	B	27	B	37	B
8	A	18	B, D	28	A, C	38	A, C
9	B	19	B	29	B	39	B
10	B	20	B	30	C	40	C

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